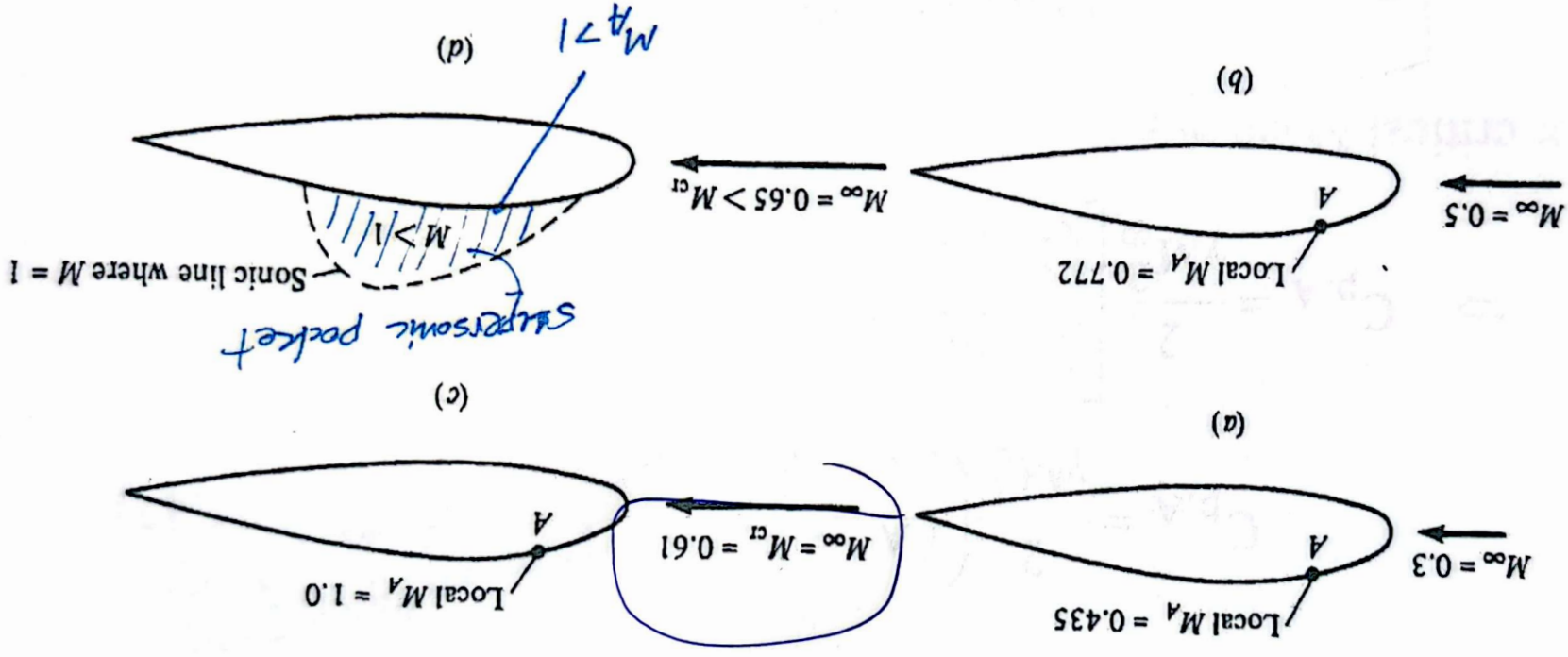


# Critical Mach number

$A = \max M \#$



$M_\infty = 0.61$  is the critical Mach number (~~#~~ Max velocity =  $M = 1$ )

if  $M_\infty > M_{cr}$  increase in drag (transonic)

For isentropic flow  $p_t$  constant

$$C_{p,A} = \frac{2}{\gamma M_\infty^2} \left( \frac{p_A}{p_\infty} - 1 \right)$$

from isentropic flow relations (8.36)

$$\frac{p_{t1}}{p_0} = \left( 1 + \frac{\gamma-1}{2} M^2 \right)^{\gamma/(\gamma-1)} \quad (5.47)$$

$$\Rightarrow C_{p,A} = \frac{2}{\gamma M_\infty^2} \left[ \left( \frac{1 + [(\gamma-1)/2] M_\infty^2}{1 + [(\gamma-1)/2] M_A^2} \right)^{\gamma/(\gamma-1)} - 1 \right] \quad (5.48)$$

for critical Mach number:  $M_A = 1$

$$\rightarrow C_{p_{cr}} = \frac{2}{\gamma M_{cr}^2} \left[ \left( \frac{1 + [(\gamma-1)/2] M_{cr}^2}{1 + (\gamma-1)/2} \right)^{\gamma/(\gamma-1)} - 1 \right] \quad (5.49)$$

Pressure coef. @ critical point

this equation can be used for all airfoils

 $M_{cr}$  $C_{p_{cr}}$ 

as  $M \uparrow$   $C_{p_{cr}} \downarrow$

0.4

- 3.66

0.6

- 1.29

0.7

- 0.78

0.75

- 0.59

0.8

- 0.43

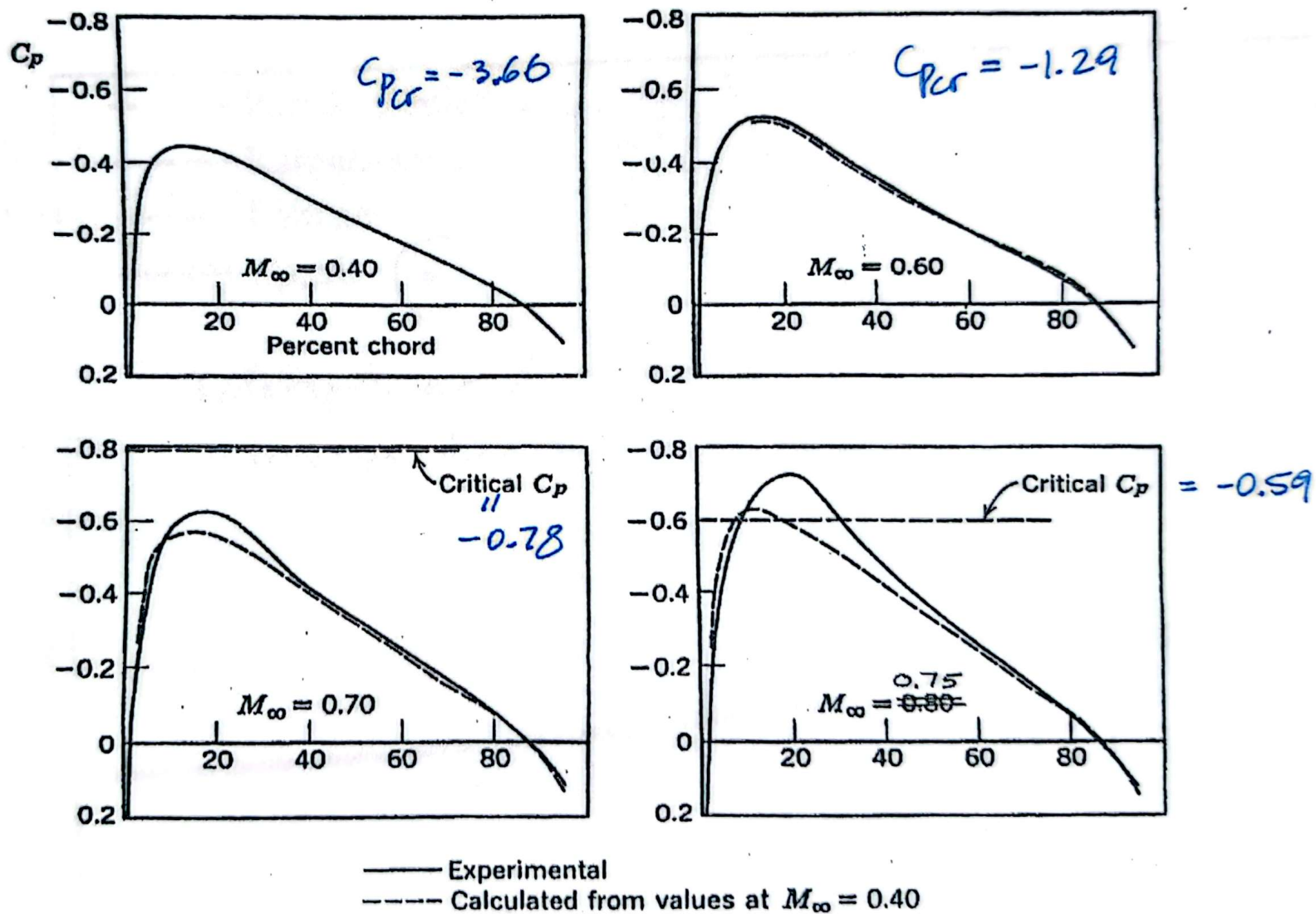
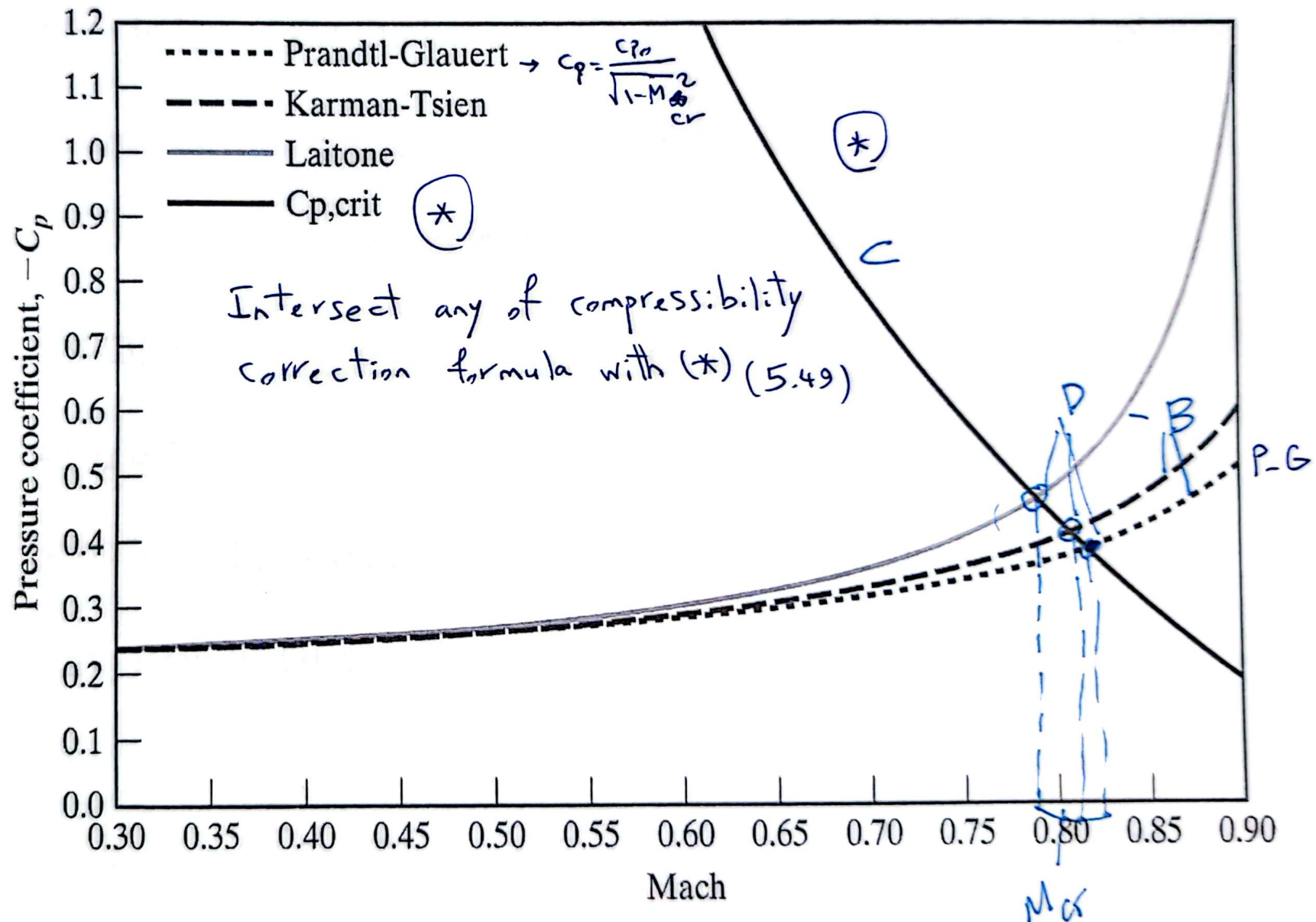


FIG. 10-1 Comparison of Prandtl-Glauert similarity rule with experiment. (Experimental data for NACA 0012 airfoil, taken from *NACA Tech. Note 2174* by J. L. Amick.)



$M_{cr}$  is lower for thicker airfoils; but needs to be high in airplanes

$\Rightarrow$  thin airfoils e.g. Gates-Lear – 9% thick

Piper Aztec 14% thick (low speed)

Example

$$c_{p_o} = c_{p'_o} = -0.25$$

At  $\alpha = 0^\circ$  the incompressible min pressure coefficient for a NACA

0009 airfoil is  $-0.25$ . Calculate the  $M_{cr}$  for this airfoil using

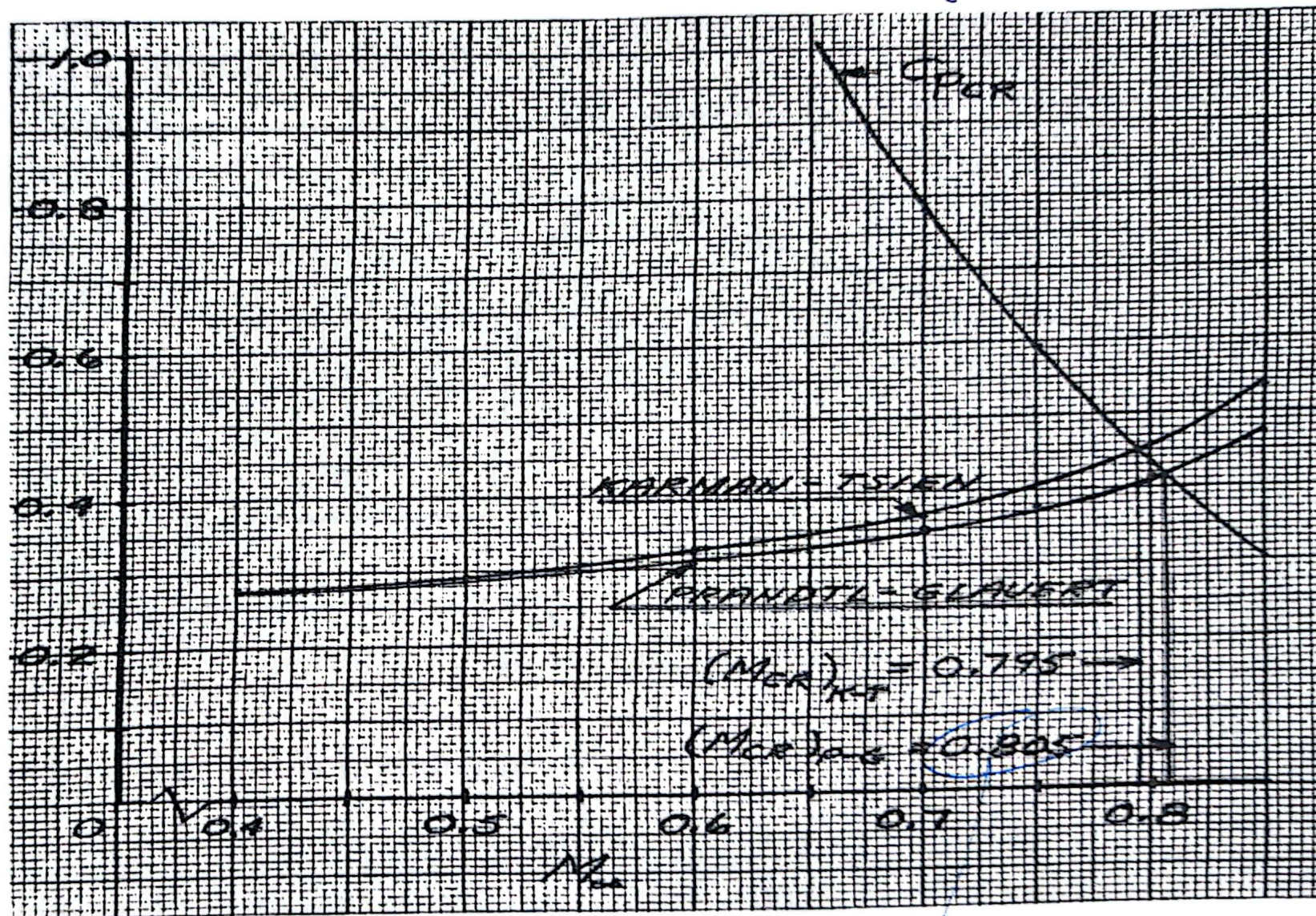
a) Prandtl-Glauert Rule  $\rightarrow c_p = \frac{c_{p_o}}{\sqrt{1-M_{cr}^2}} = \frac{-0.25}{\sqrt{1-M_{cr}^2}} \quad (1)$

b) The (more accurate) Karman-Tsien rule  $K.T \text{ formula } (2)$

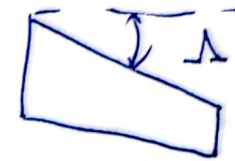
To solve  $\begin{cases} (1) \text{ write a MATLAB/Python code that solve } (1 - (*)^2) = 0 \\ (2) \text{ plot } (1) \text{ and } (*) \text{ in terms of Mach No. and find the intersection point.} \end{cases}$

| $M_\infty$ | $C_{p_{cr}}$ eq. (9.21) | $C_{p_{Pr andtl-Glauret}}$ | $C_{p_{Karman-Tsien}}$ |
|------------|-------------------------|----------------------------|------------------------|
| 0.3        | - 6.9                   | - 0.262                    | - 0.264                |
| 0.4        | - 3.66                  | - 0.273                    | - 0.276                |
| 0.5        | - 2.13                  | - 0.289                    | - 0.294                |
| 0.6        | - 1.29                  | - 0.313                    | - 0.323                |
| 0.7        | - 0.78                  | - 0.35                     | - 0.369                |
| 0.75       | - 0.59                  | - 0.378                    | - 0.404                |
| 0.80       | - 0.43                  | - 0.417                    | - 0.455                |
| 0.85       | - 0.302                 | - 0.475                    | - 0.535                |

(x)

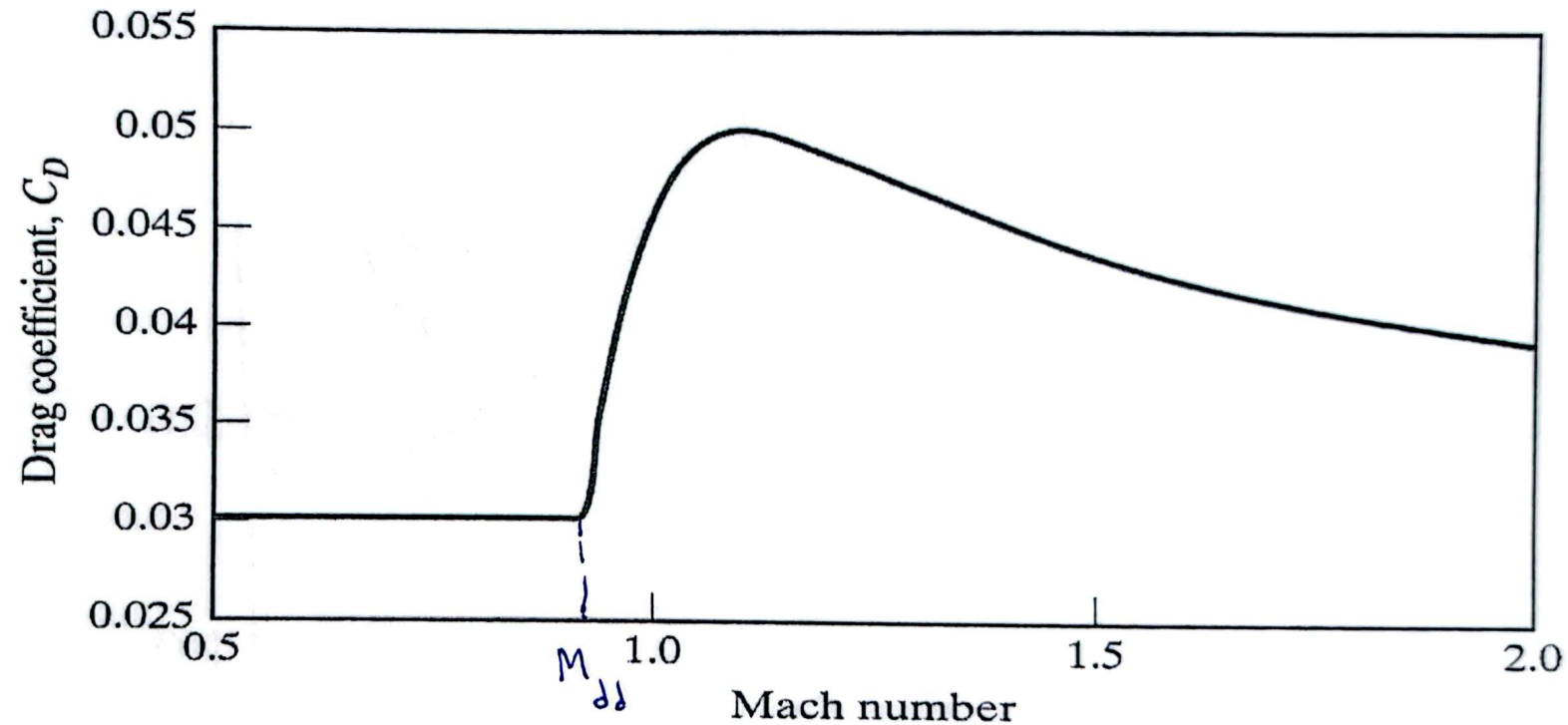


a little higher



Sweep angle <sup>29</sup>

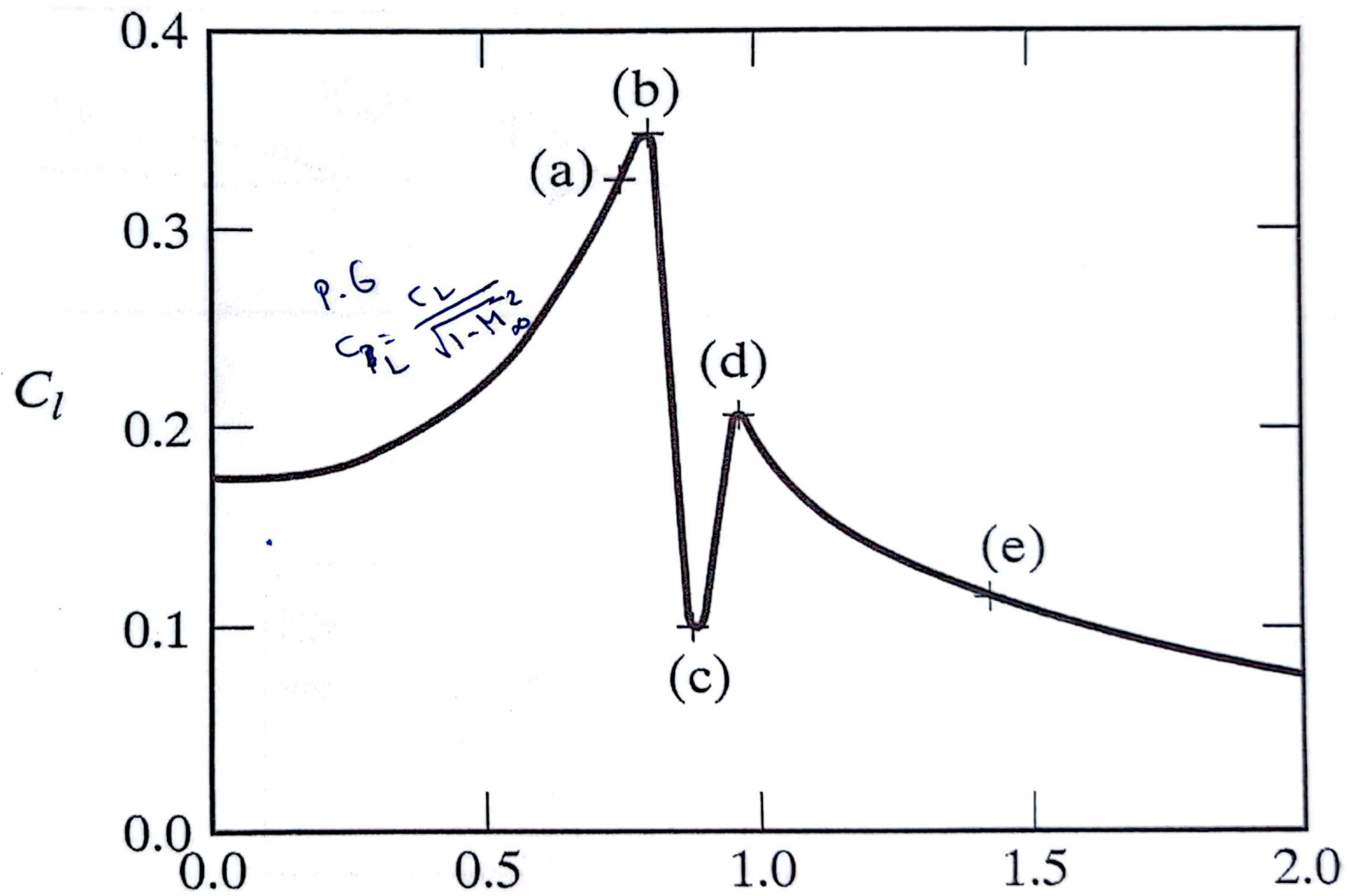
## Drag Divergence Mach Number



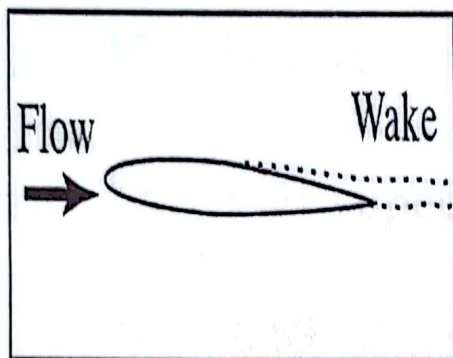
$M_{dd}$  either 2-4% higher than  $M_{cr}$ , or  $dC_D / dM_\infty = 0.03, 0.05, 0.10$

$$M_{dd} = \underbrace{M_{cr}}_{\text{sweep angle}} \left[ 1.02 + 0.08(1 - \cos \Lambda) \right] \quad (9.22)$$

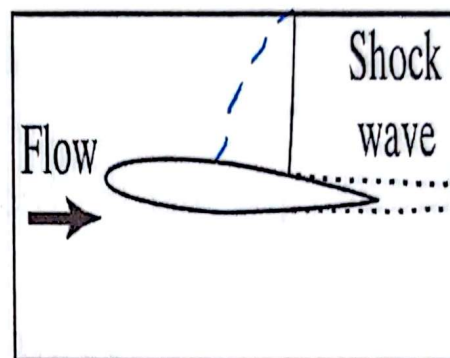
empirical approximation based on  $dC_D / dM_\infty = 0.05$



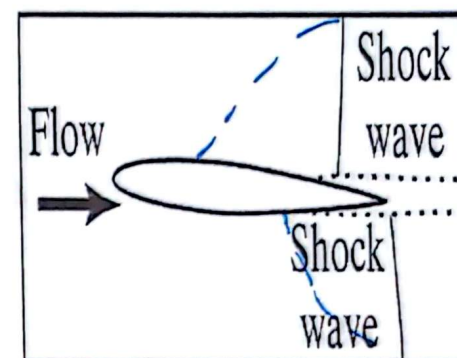
lift coefficient as a function of  $M$



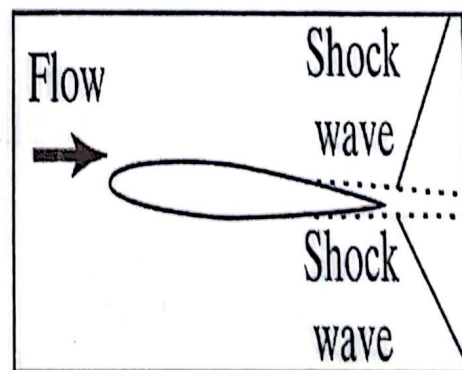
(a)



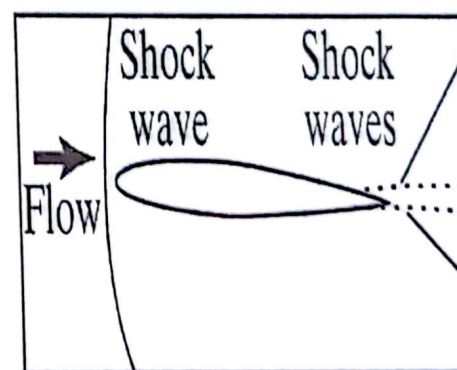
(b)



(c)



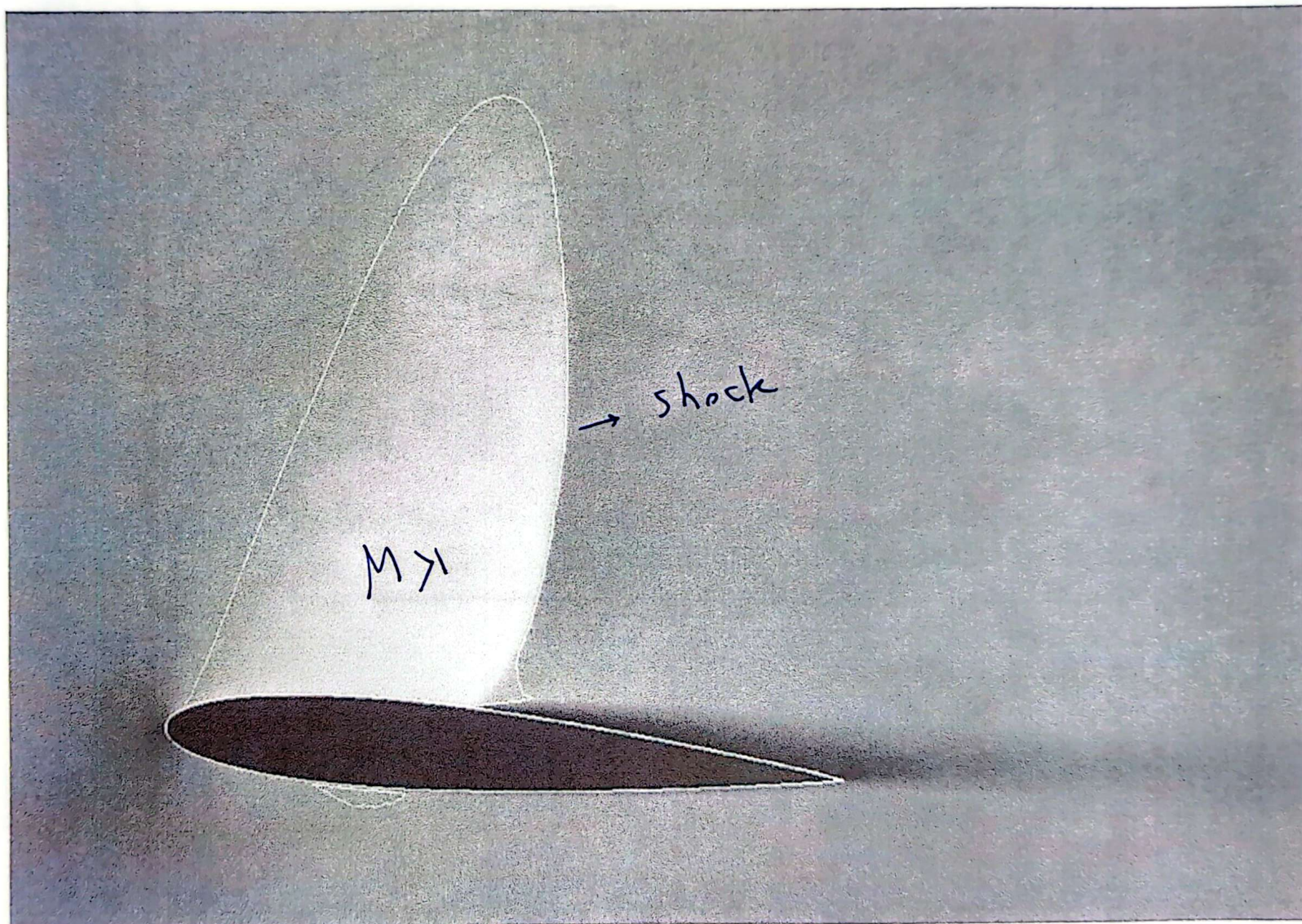
(d)



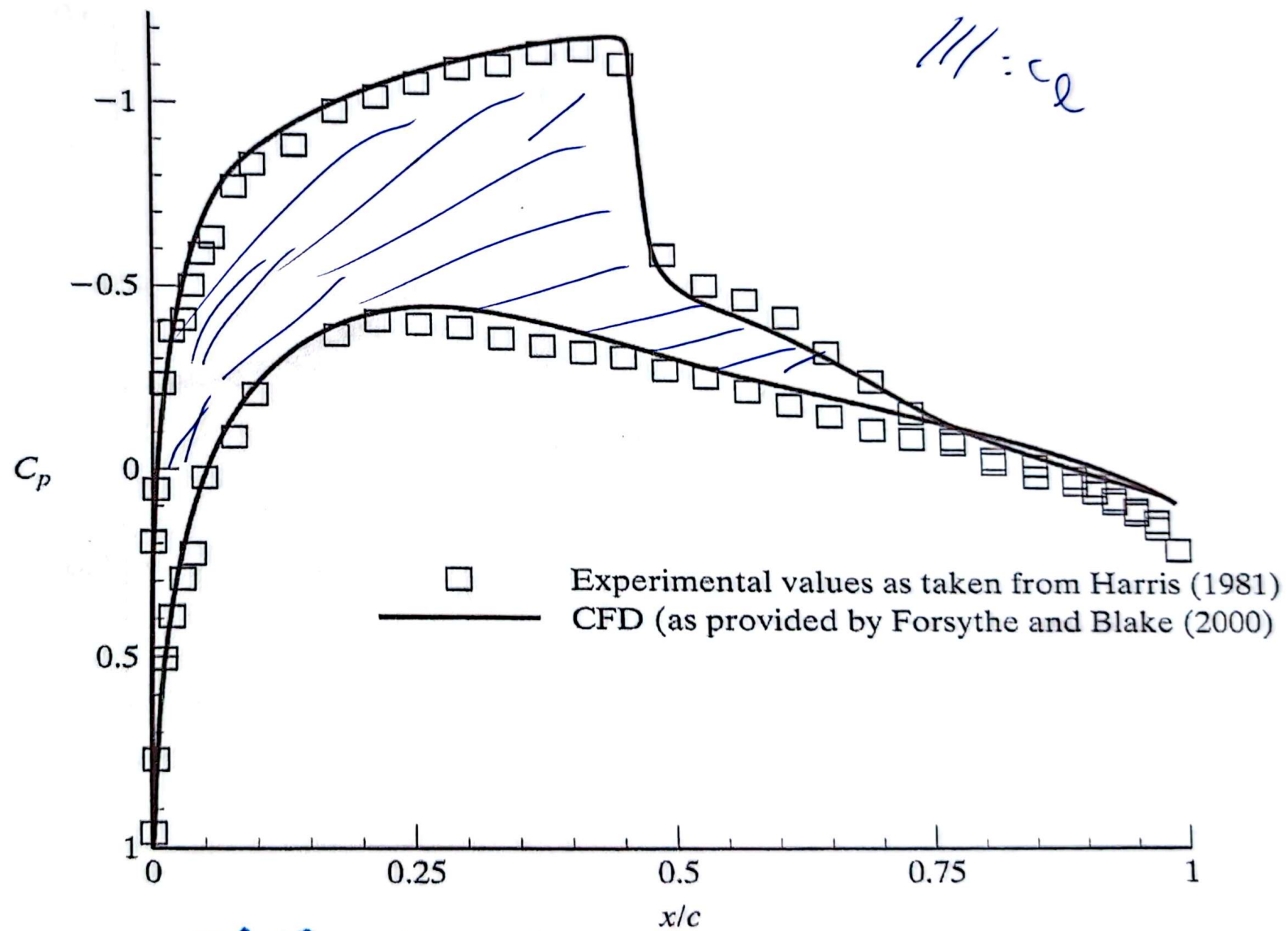
(e)

increasing  $M$ :

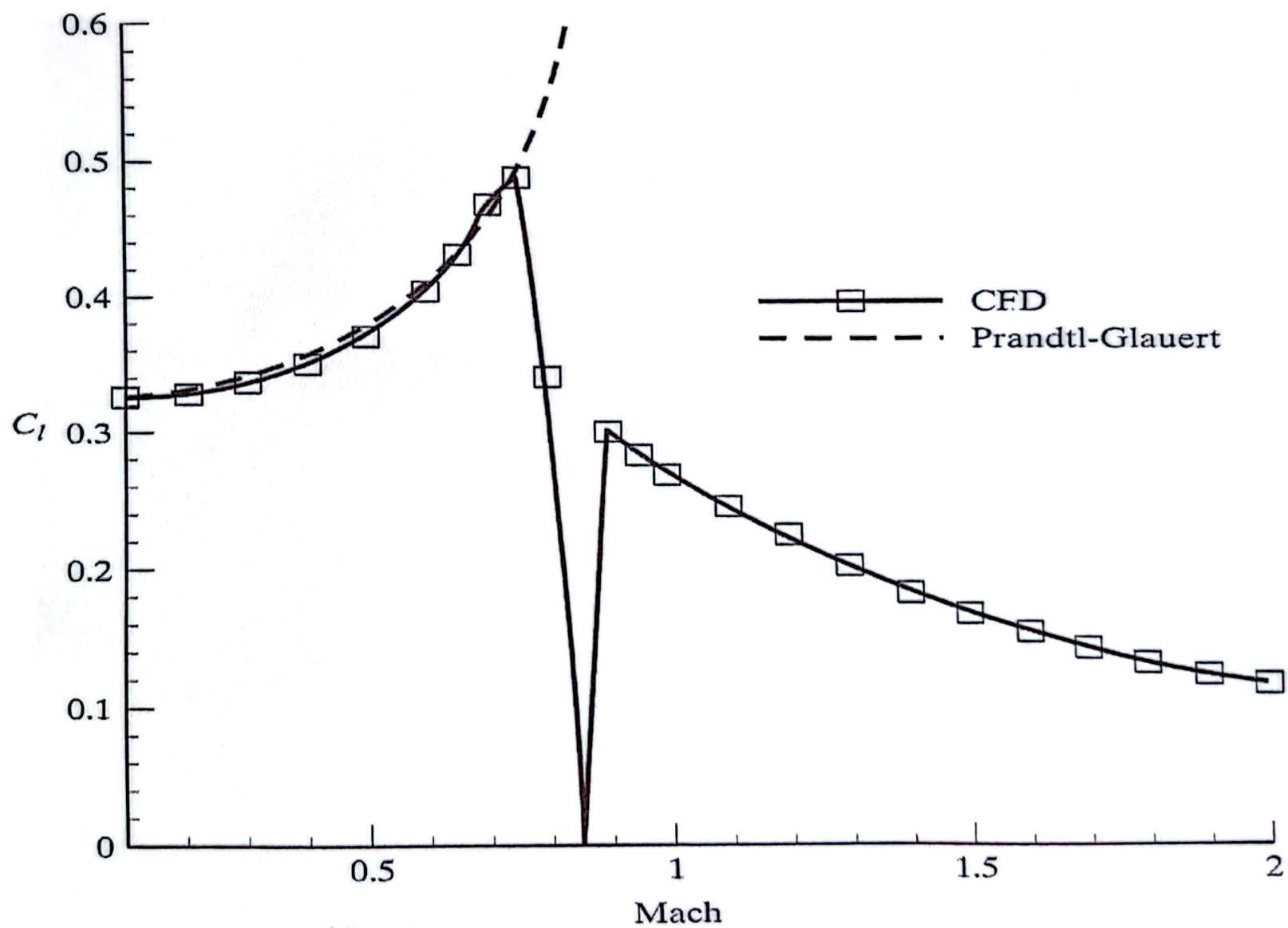
- a)  $M_\infty = 0.75$  subcritical
- b)  $M_\infty = 0.81$  supercritical on the top, shock wave maximum lift
- c)  $M_\infty = 0.89$  shock in upper surface becomes stronger; shock in lower surface
- d)  $M_\infty = 0.98$  both shocks reach trailing edge
- e)  $M_\infty > 1$  detached shock



NACA 0012,  $M = 0.8$ ,  $\alpha = 3^\circ$



NACA 0012,  $M=0.8$ ,  $\alpha=3^\circ$

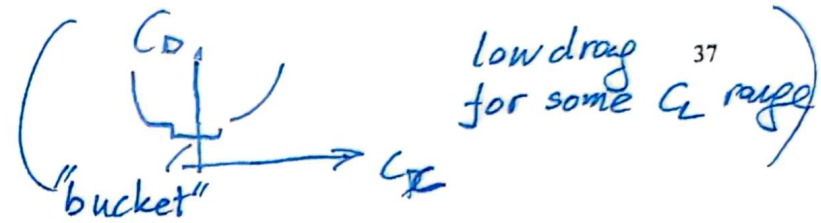


NACA 0012



B-1B transonic flight (condensation & evaporation of water)

## Supercritical airfoils



- i) increase  $M_{cr} \rightarrow$  thin airfoils (1945-1965) NACA 64 series  
(originally developed to encourage laminar flow) need thick airfoil for: strength - fuel storage
- ii) increase the Mach number increment between  $M_{cr}$  and  $M_{dd}$

1945-1965- NACA 64 series

1965  $\rightarrow$  now  $\Rightarrow$  supercritical airfoils

*used in the 80's*

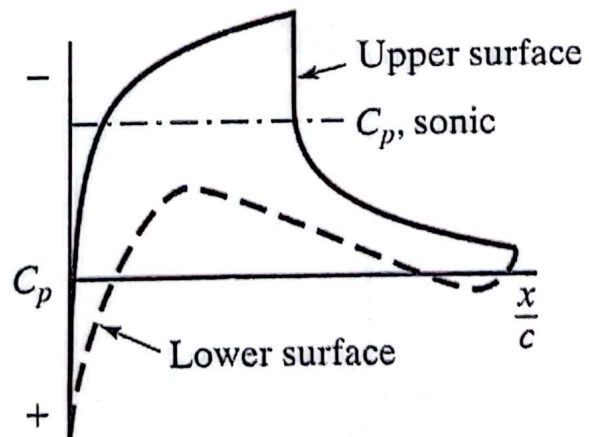
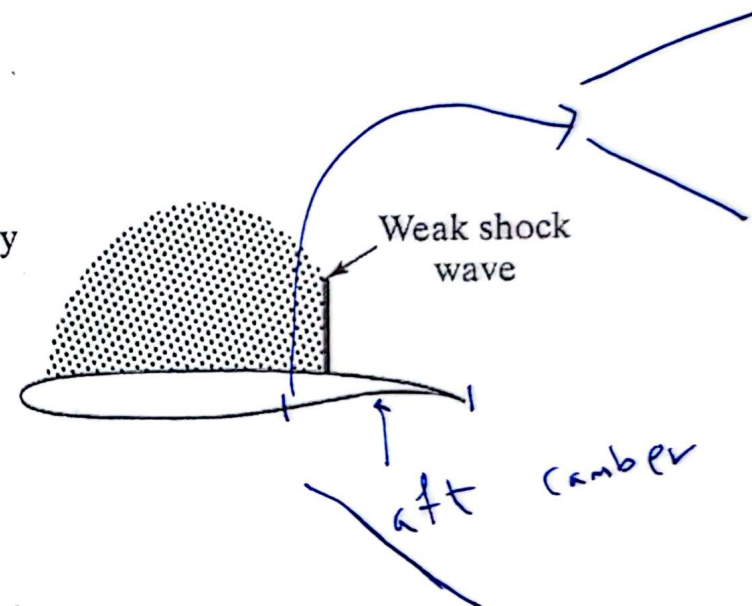
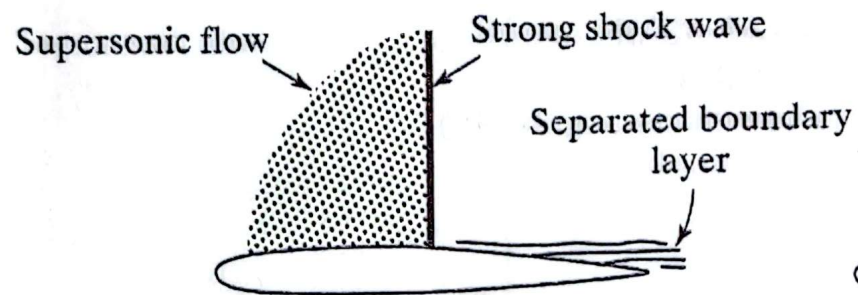
Regular airfoils have a strong shock (separation, high drag)

Supercritical airfoils have a flatter top; shock further back, weaker, mild or no separation.

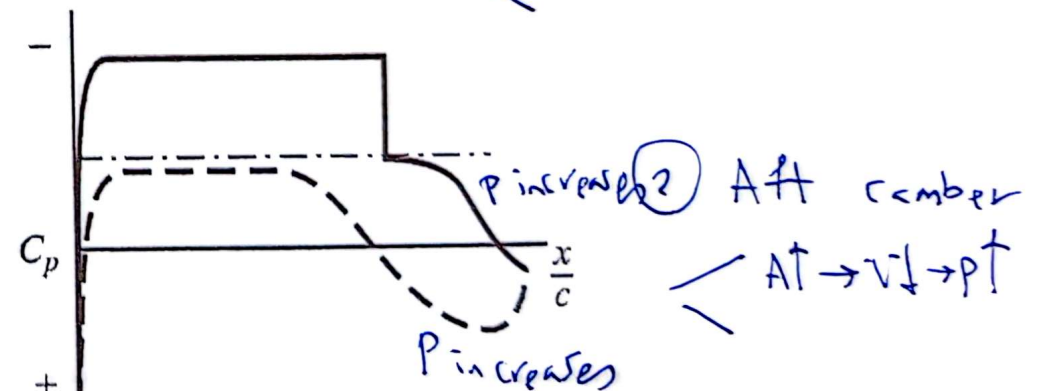
Off-design characteristics suffer lift and drag penalties *(later release)*

*Question: 757, 767, 777 use supercritical airfoils; 717 did not. why?*

*Answer: cost (used earlier wing from an MD airplane)*



(a)  
ordinary



(b)  
super-critical { ① straight upper surface

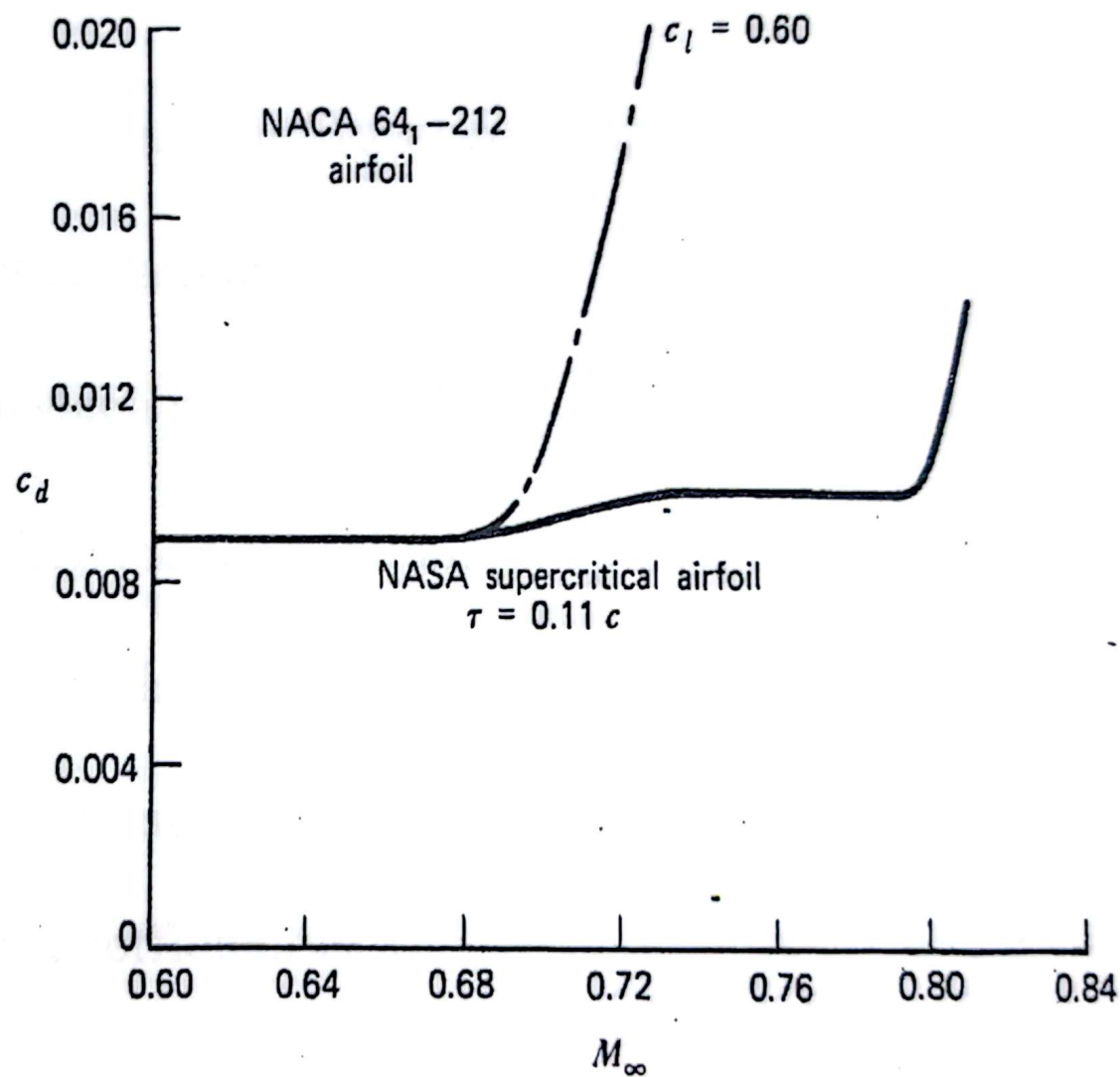


Fig. 8. Characteristics of NASA supercritical airfoil compared with a conventional section. (Whitcomb, 1956. *Courtesy of NASA*)

## Swept Wings

Independence Principle: flow can be split in normal + tangential

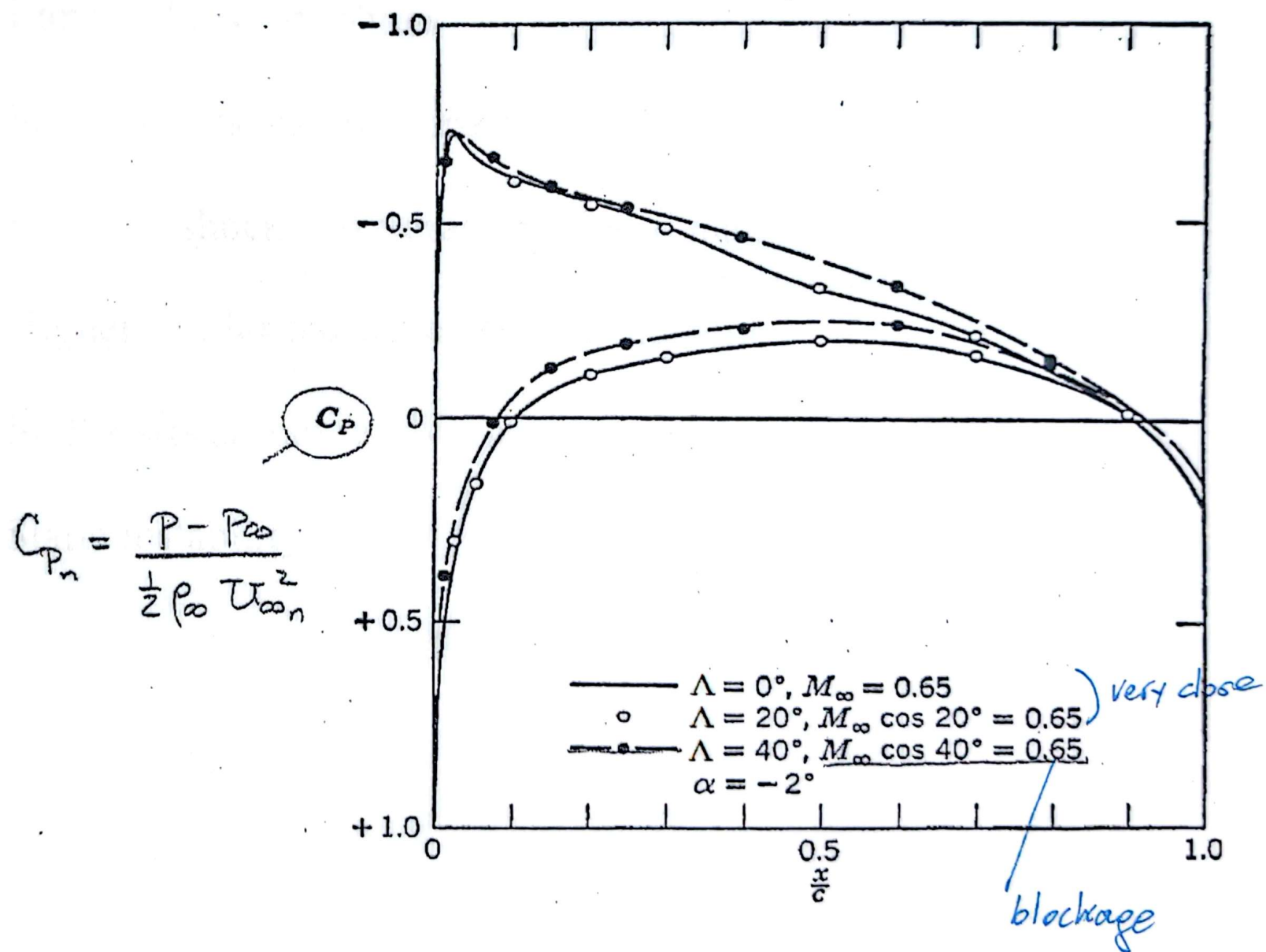
Assume a sweep angle  $\Lambda$

The normal Mach number is:  $M_{n, \infty} = M_{\infty} \cos \Lambda$

The point when the flow becomes supersonic or goes above drag divergence is delayed; any shock is weaker or delayed

The critical normal Mach number:  $M_{n, cr} = M_{\infty, cr} \cos \Lambda$

The pressure coefficient is:  $C_{p n} = \frac{p - p_{\infty}}{\frac{1}{2} \rho_{\infty} U_{\infty}^2 \cos^2 \Lambda} = \frac{\frac{p}{p_{\infty}} - 1}{\frac{1}{2} \gamma M_{n, \infty}^2}$



## Forward Swept Wings

Structural issues; use aeroelastic tailoring and composites

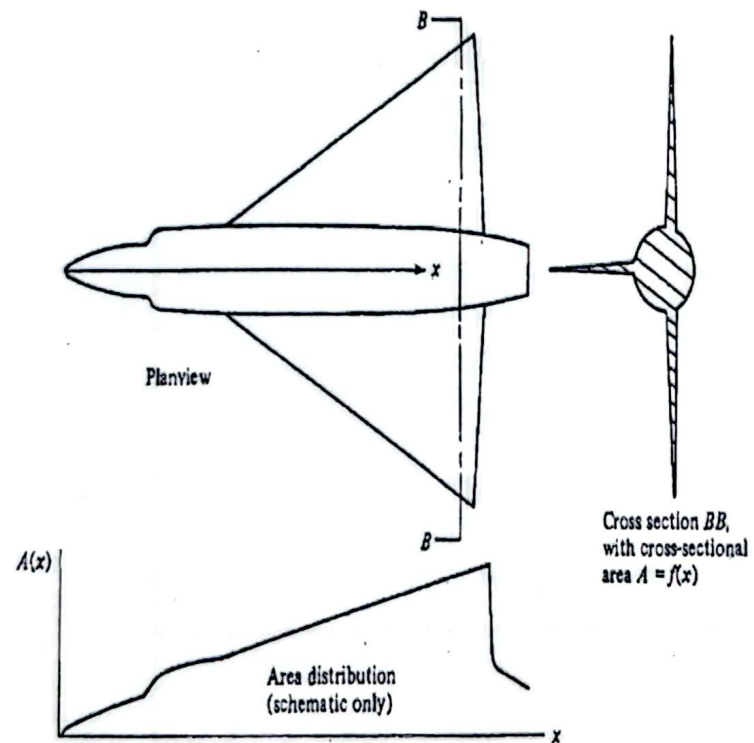
Reduced shock wave drag, increased maneuverability  
*(important for fighter airplanes)*

Higher AR for the same bending moment

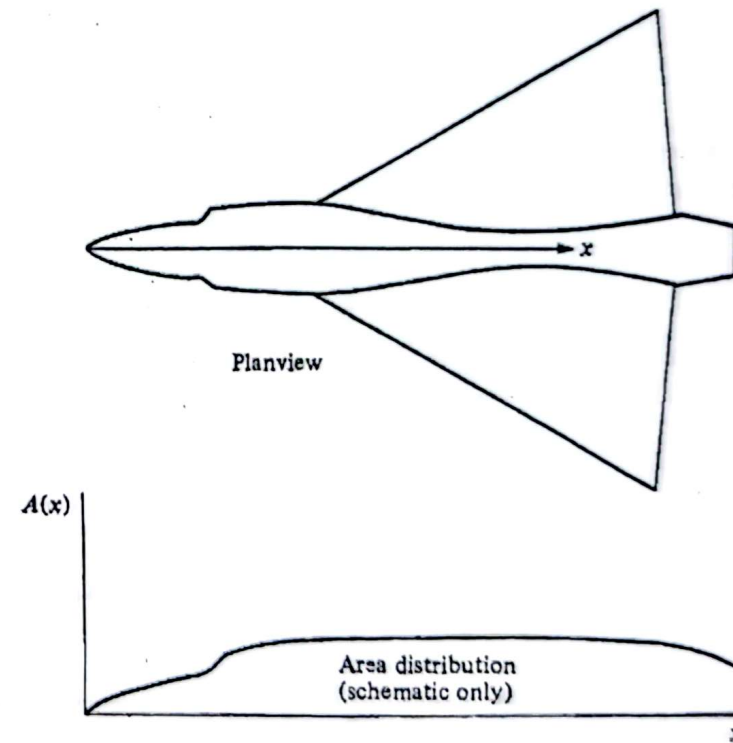
Stall starts at root (early warning, maintain aileron effectiveness)

Static instability

## Transonic Area Rule: Area varies smoothly

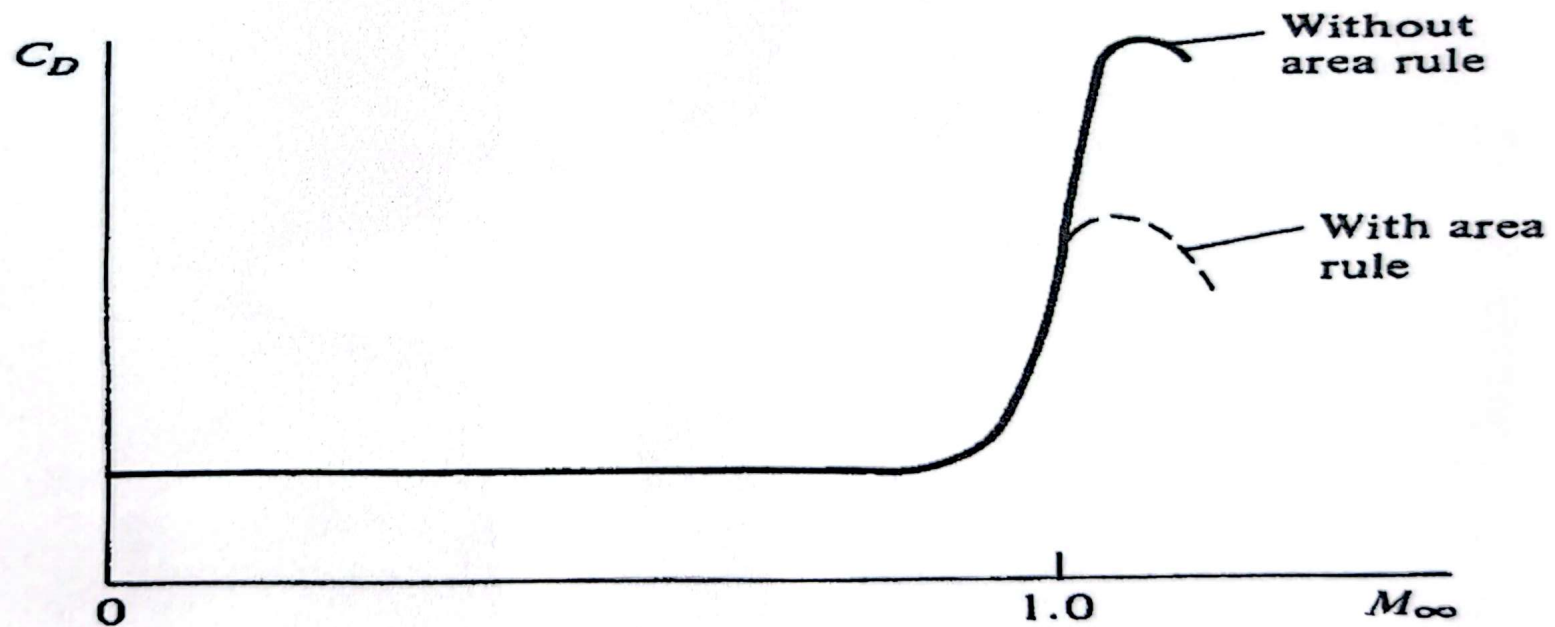


**FIGURE 11.10**  
A schematic of a non-area-ruled aircraft.

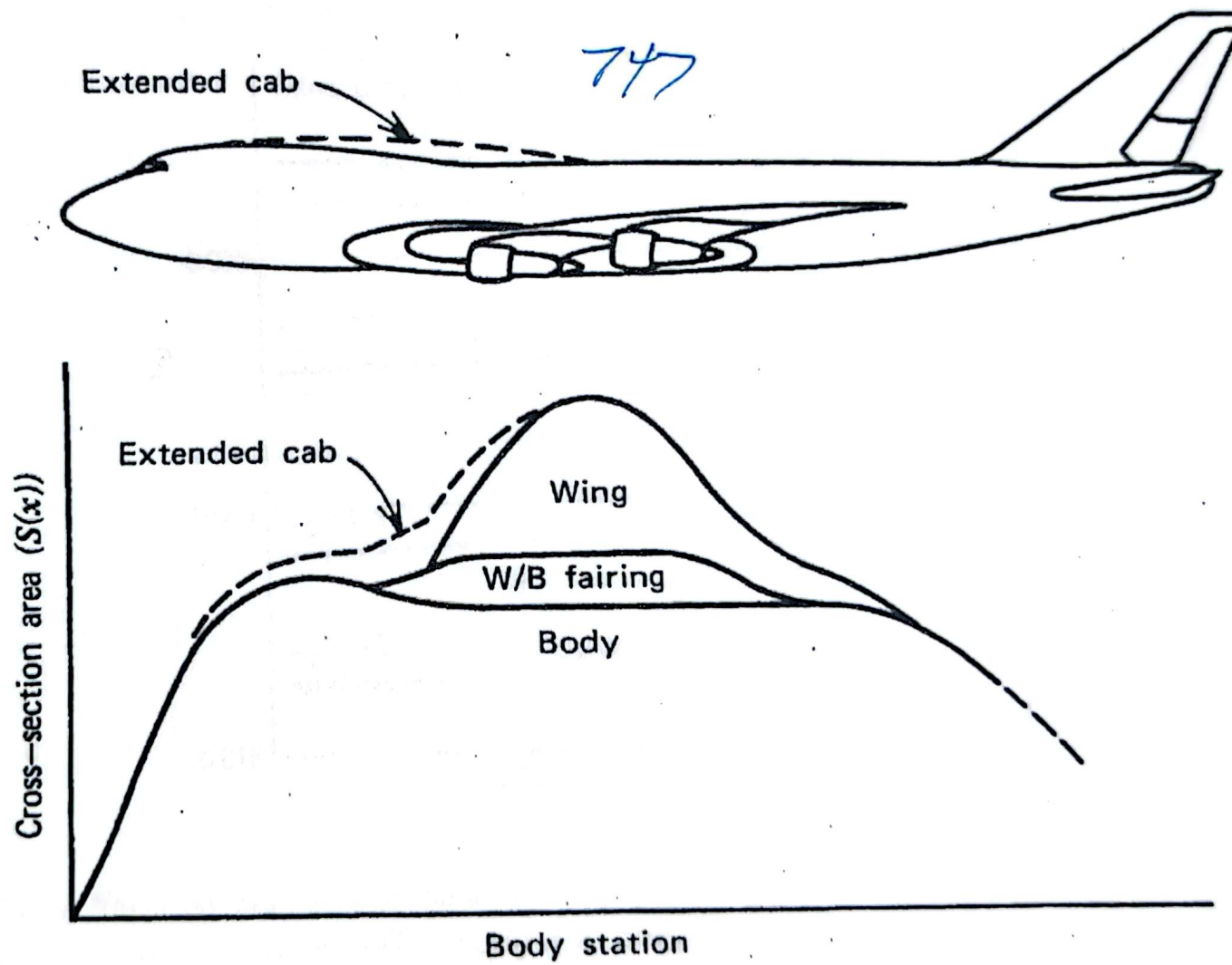


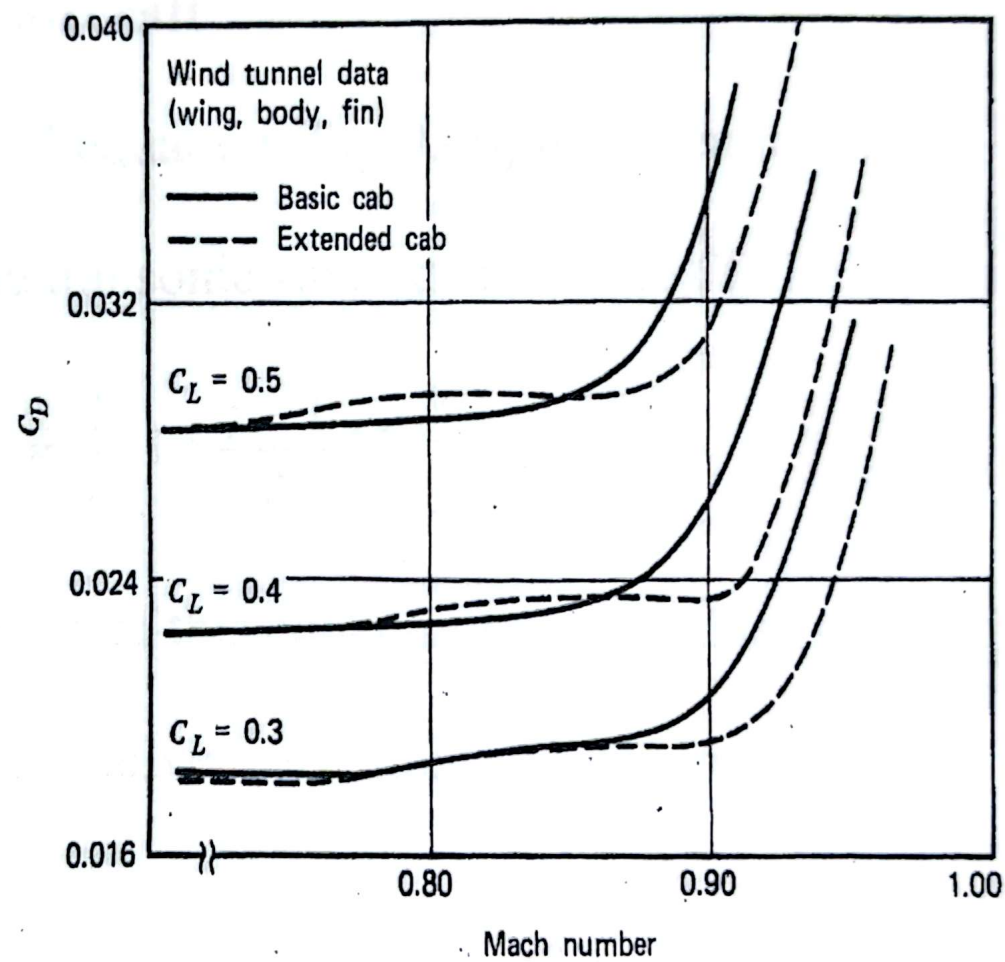
**FIGURE 11.11**  
A schematic of an area-ruled aircraft.

ballistics: speed of supersonic bullets is better for smooth variation of cross-sectional area



Whitcomb developed the rule in the mid 1950's





**Fig. 9.** Effect on  $S(x)$  and on measured drag of Boeing 747 due to fuselage modification. (Goodmanson and Gratzner, 1973. *Courtesy of the Boeing Company*)

## Transonic aircraft

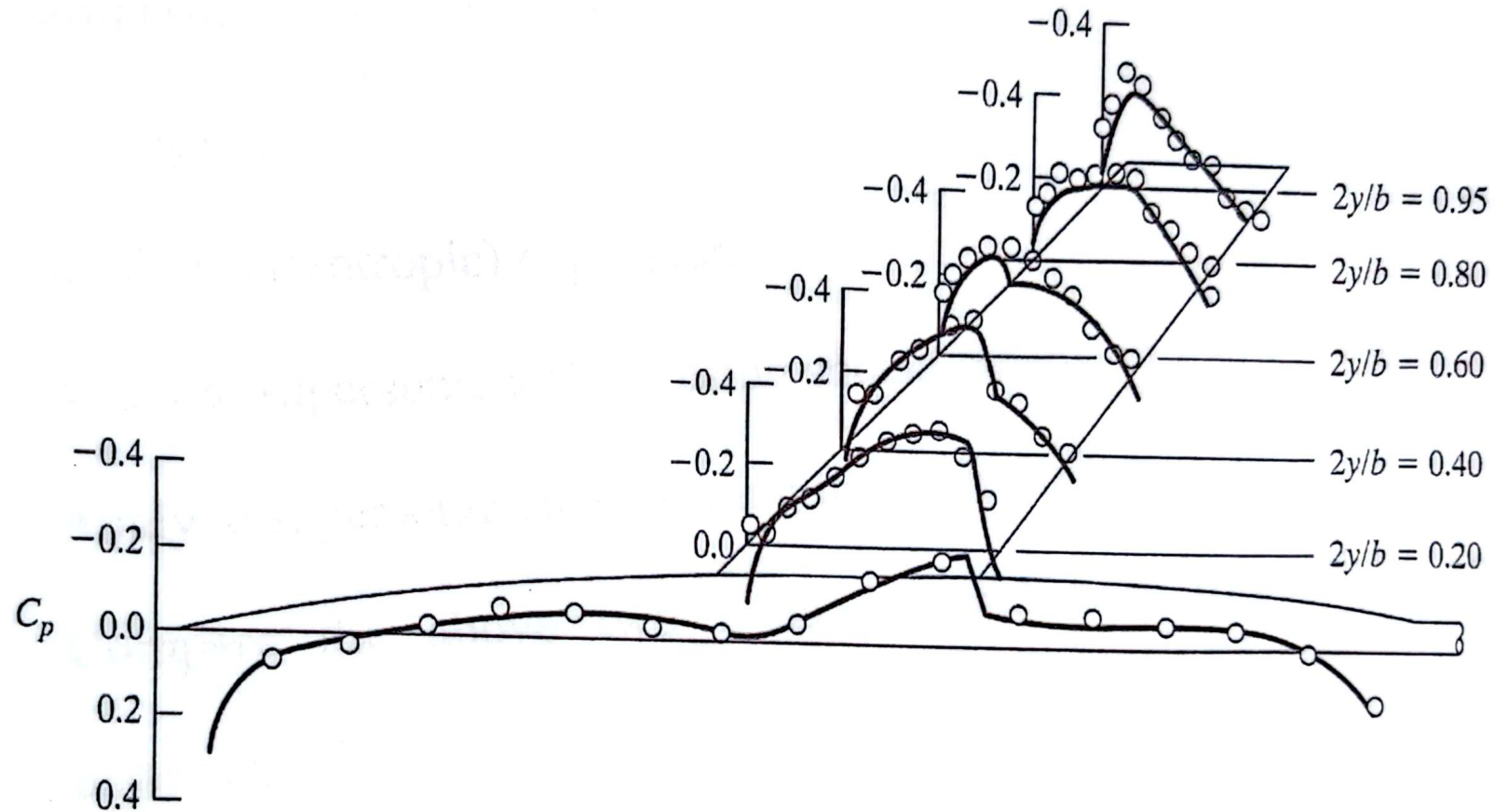
Full potential equation (Eq. 9.4b) has been used. *extensively*  
*(flow over wings)*  
 Faster is the transonic small disturbance (TSD) equation

$$\beta^2 \phi_{xx} + \phi_{yy} = \frac{2}{\gamma} \left( 1 + \frac{\gamma-1}{2} M_\infty^2 \right) M_\infty^2 \phi_x \phi_{xx} \quad (9.23)$$

Other versions of this equation are available (see text)

*Euler &*

Navier Stokes solvers are increasingly used.



$C_p$  for swept-wing/fuselage configuration,  $M_\infty = 0.93$ ;  $\alpha = 0^\circ$ ,  $\Lambda_{c/4} = 45^\circ$ ,  $AR = 4$ ,  $\lambda = 0.6$ , NACA 65A006 streamwise section.

small disturbance eq. Bailey Bellhaus 1975

## **Supersonic Flow around Airfoils [8.5, Chapter 10]**

---

### **Objective:**

Analyze (isentropic) expansion

Analyze supersonic airfoils using shock/expansion wave patterns

Analyze supersonic airfoils using linearized theory

Compare the above two methods (including limitations and assumptions)

Euler's equation:  $dp = -\rho U dU$

From (8.56a)  $\frac{dU}{U} = \frac{d\theta}{\sqrt{M^2 - 1}}$  (8.57)

*adiabatic flow*  $\left(\frac{U}{a_i}\right)^2 = M^2 \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1}$  (8.58)

Differentiating and substituting to (8.57)

$$d\theta = \frac{\sqrt{M^2 - 1} dM^2}{2M^2 \{1 + [(\gamma - 1)/2] M^2\}} \quad (8.59)$$

Integrating:  $\theta = \nu + \text{constant}$ ;  $\nu$  is the Prandtl-Meyer function

$$\nu = \sqrt{\frac{\gamma + 1}{\gamma - 1}} \tan^{-1} \sqrt{\frac{\gamma - 1}{\gamma + 1} (M^2 - 1)} - \tan^{-1} \sqrt{M^2 - 1} \quad (8.60)$$

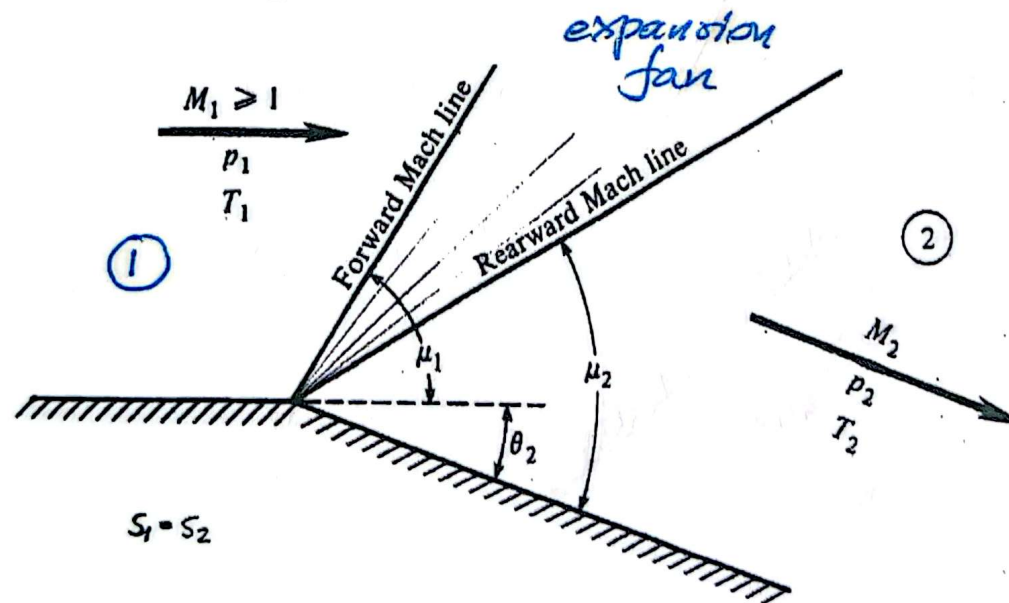
Note:  $\nu(M) = 0$  when  $M = 1$

$\nu$  is tabulated as a function of  $M$  (Table 8.3)

Along left running characteristics:  $v - \theta = R$

Along right running characteristics:  $v - \theta = Q$

Example – 8.3 Supersonic thin airfoil [see text] – *not important*  
 For a finite expansion (for left running characteristics; top surface),  
 isentropic flow  $\Delta v = \Delta \theta$



Process:

find  $v(M_1)$  from Table 8.3

find  $v(M_2)$  from

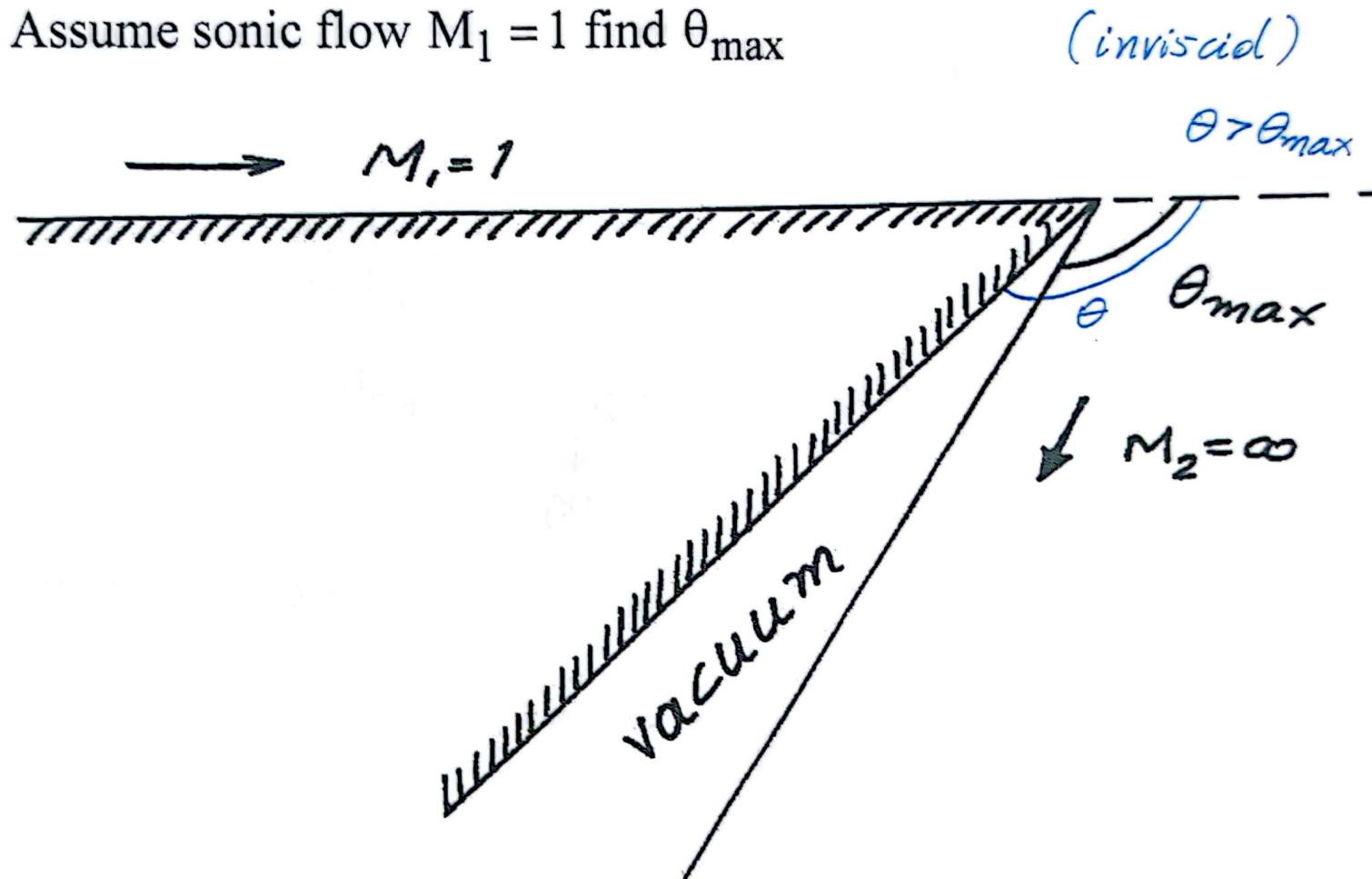
$$\theta = v(M_2) - v(M_1)$$

find  $M_2$  from Table 3

find  $p_2, T_2$  for isentropic  
 flow (Table 8.1)

$$M_2 > M_1 \quad ; \quad p_2 < p_1$$

Assume sonic flow  $M_1 = 1$  find  $\theta_{\max}$



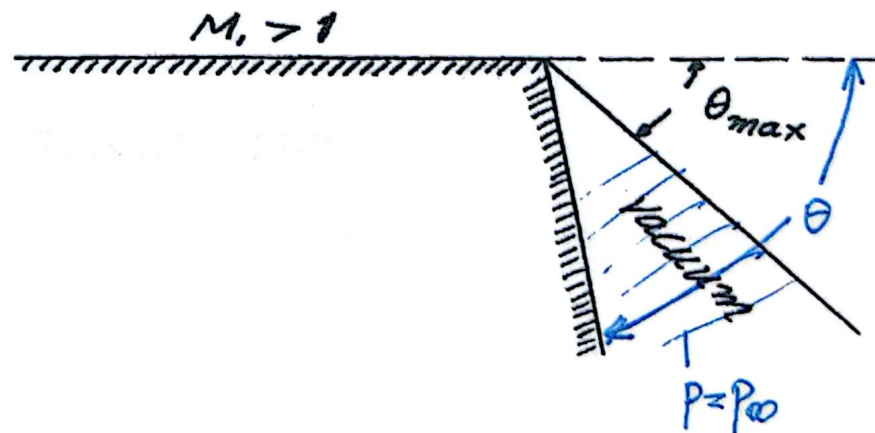
$$\lim_{M_2 \rightarrow \infty} v_2 = \lim \left( \sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1} \sqrt{\frac{\gamma-1}{\gamma+1} M_2^2 - 1} - \tan^{-1} \sqrt{M_2^2 - 1} \right)$$

$$= \sqrt{\frac{\gamma+1}{\gamma-1}} \frac{\pi}{2} - \frac{\pi}{2} = \left( \sqrt{\frac{\gamma+1}{\gamma-1}} - 1 \right) \frac{\pi}{2} = 2.277 \text{ rad} = 130^\circ.45$$

$$\theta_{\max} = v(M_2) - v(M_1)$$

$\downarrow$                        $\downarrow$   
 $130^\circ.45$               0 for  $M_1 = 1$

for any flow a  $\theta_{\max}$  can be found e.g.

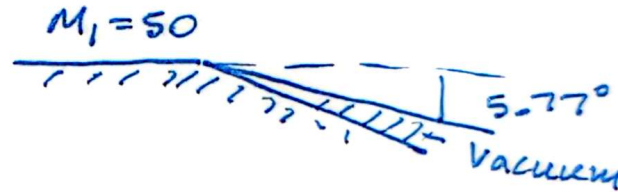


*tables*

$$M_1 = 5 \rightarrow v(M_1) = 76^\circ.92 \Rightarrow \theta_{\max} = 53^\circ.53$$

$$M_1 = 50 \Rightarrow v(M_1) = 124^\circ.7 \Rightarrow \theta_{\max} = 5^\circ.77$$

as



$$M_1 \uparrow \Rightarrow \theta_{\max}$$

$$M_1 \rightarrow \infty \Rightarrow \theta_{\max} \rightarrow 0$$

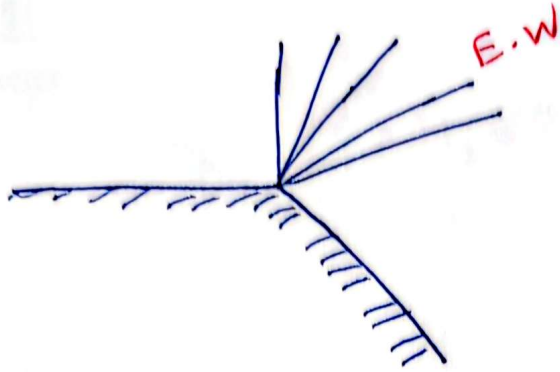
*no expansion*



(flow cannot accelerate anymore)

- see Newtonian theory for hypersonic flow  
ch. 12

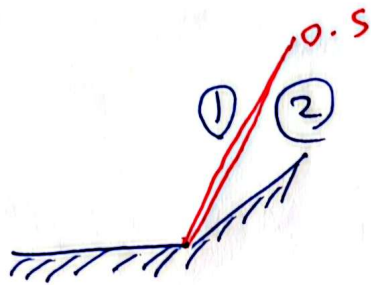
Expansion waves : When there is area increase, there are expansion waves



E.W are Mach waves

$$E.W \rightarrow \begin{cases} P_2 < P_1 \\ M_2 > M_1 \\ T_2 < T_1 \text{ and } s_2 < s_1 \end{cases}$$

oblique shock: when area decreases



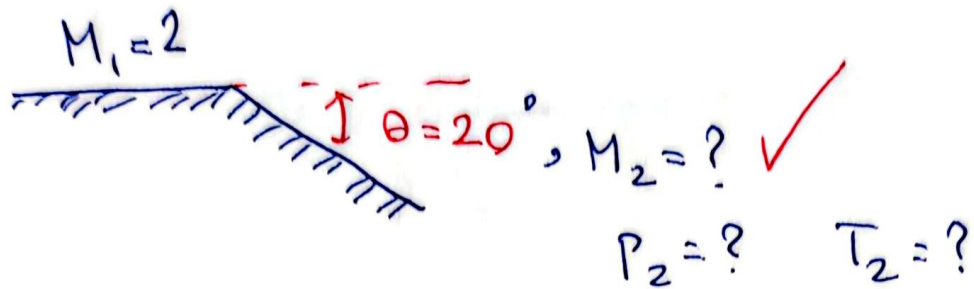
$$Shock \begin{cases} P_2 > P_1 \\ M_2 < M_1 \\ T_2 > T_1 \\ s_2 > s_1 \end{cases}$$

"E.W and O.S acts oppositely"

E.W Formula:  $\underbrace{\nu(M_2)} - \nu(M_1) = \theta_2 - \theta_1$

A function  
of Mach

Example:



Solution:

$$\nu(M_2) - \nu(M_1) = \underbrace{\theta_2}_{20^\circ} - \underbrace{\theta_1}_{0^\circ} \quad (\text{use degree})$$

$$M_1 = 2 \xrightarrow[\text{Calc}]{\text{Isentropic online}} \text{read } \nu(M_1) = 26.379 \quad \begin{cases} \frac{P_1}{P_0} = 0.127 \\ \frac{T_1}{T_0} = 0.55 \end{cases}$$

$$\Rightarrow \nu(M_2) = \nu(M_1) + 20 - 0 = 26.379 + 20 = 46.379$$

We have  $\nu(M_2)$ , go to online calc, find  $M_2$

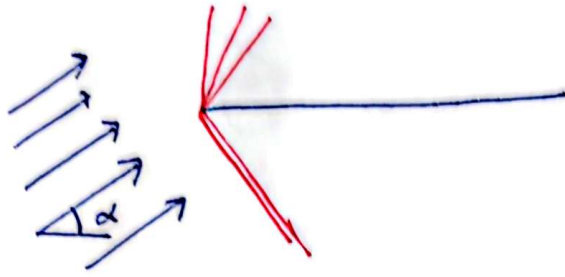
$$\nu = 46.379 \xrightarrow[\text{calc}]{\text{Online}} \begin{cases} M_2 = 2.83 \\ \frac{P_{02}}{P_0} = 0.035 \end{cases}, \quad \frac{T_2}{T_0} = 0.384$$

$$P_2 = \frac{P_2}{P_0} \times \frac{P_0}{P_1} \times P_1 = 0.035 \times \frac{1}{0.127} \times P_1 = \checkmark$$

$$T_2 = \frac{T_2}{T_0} \times \frac{T_0}{T_1} \times T_1 = 0.384 \times \frac{1}{0.55} \times T_1 = \checkmark$$

Note: "E.w are isentropic"

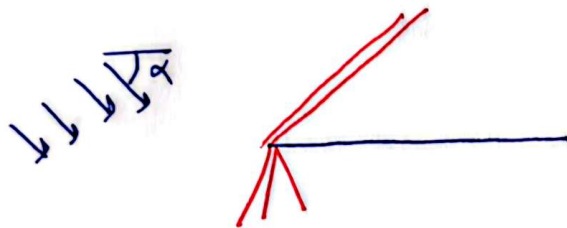
Flat plate



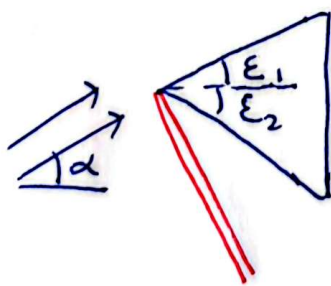
"positive lift"



Nothing happens!  
No lift



"Negative lift"



For this case, always on the bottom surface we have D.S with turn angle  $\alpha + \epsilon_2$

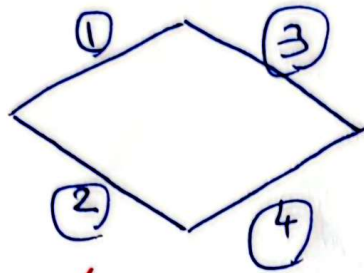
On the top surface, 3 cases may happen:

If  $\alpha = \epsilon_1$  : Nothing!

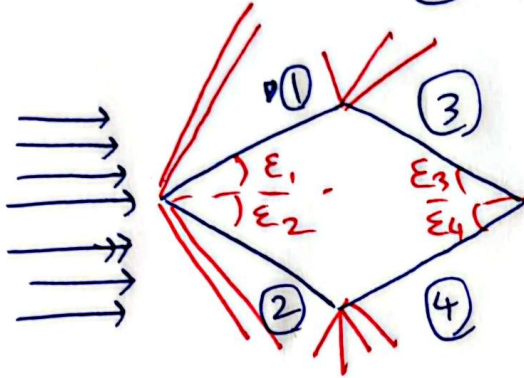
If  $\alpha > \epsilon_1$  : E.W with turn angle  $\alpha - \epsilon_1$

If  $\epsilon_1 > \alpha$  : D.S " " "  $\epsilon_1 - \alpha$

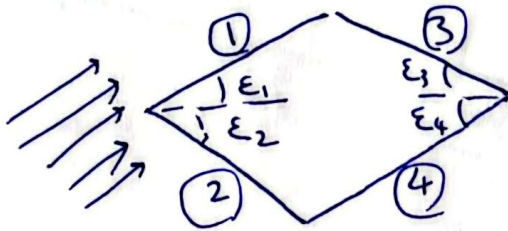
# Diamond airfoil



$\alpha = 0$



$\alpha > 0$



1-3  $\rightarrow$  E.W  
2-4  $\rightarrow$  E.W

As flow area increases, turn angle

$$\begin{cases} (\epsilon_1 + \epsilon_3) : 1-3 \\ (\epsilon_2 + \epsilon_4) : 2-4 \end{cases}$$

For every  $\alpha$   
this is the case

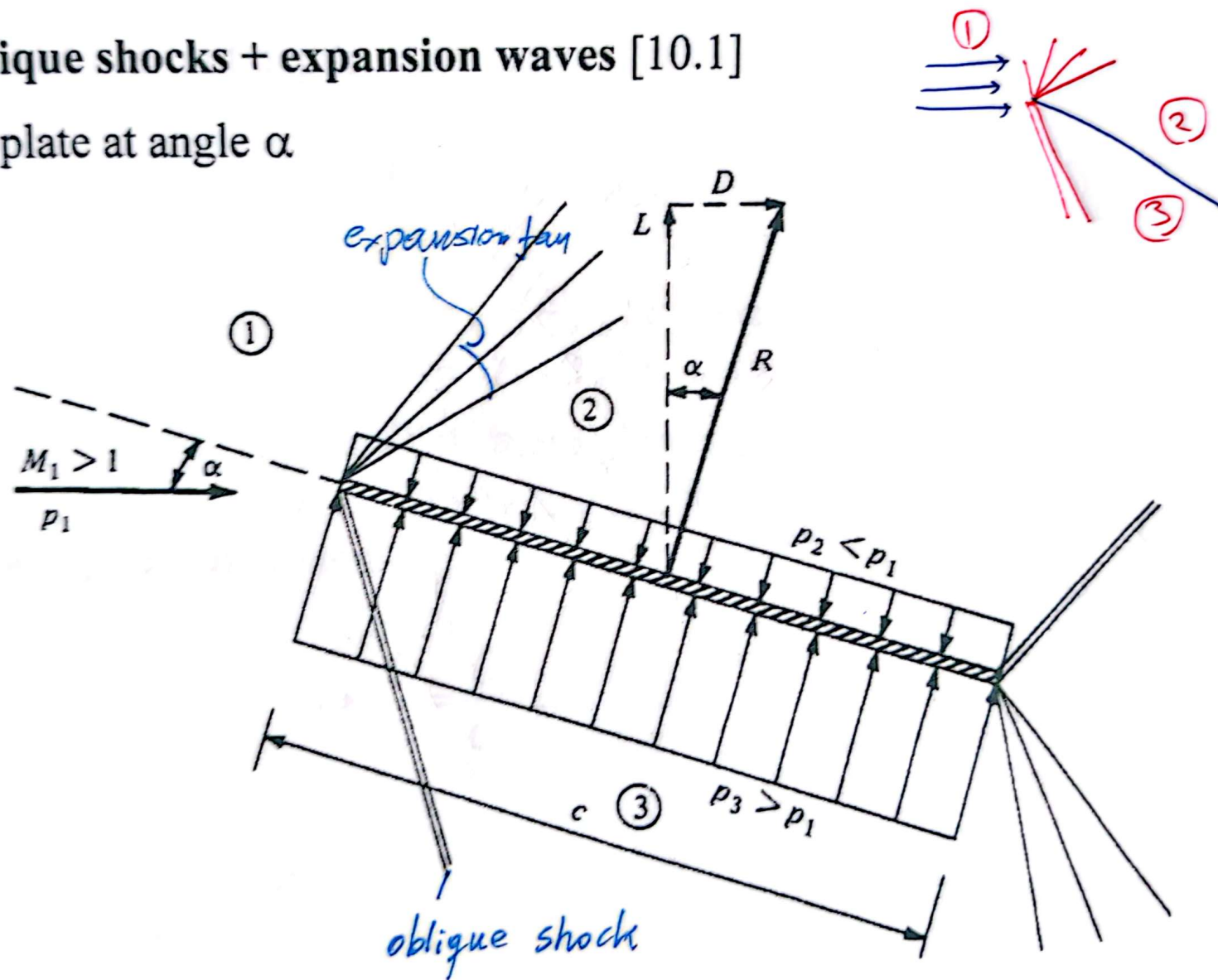
$\alpha = \epsilon_1 \rightarrow$   $\begin{cases} \text{Nothing on } ① \\ 0.5 \text{ on } ②, \text{ turn angle } \alpha + \epsilon_2 \end{cases}$   
 $s_0 \ s_0 \leftarrow$

$\alpha < \epsilon_1 \rightarrow$   $\begin{cases} 0.5 \text{ on } ①, \text{ turn angle } \epsilon_1 - \alpha \\ 0.5 \text{ on } ②, \text{ turn angle } \epsilon_1 + \alpha \end{cases}$   
Worst lift  $\leftarrow$

$\alpha > \epsilon_1 \rightarrow$   $\begin{cases} \text{E.W on } ①, \text{ turn angle } \alpha - \epsilon_1 \\ 0.5 \text{ on } ②, \text{ turn angle } \alpha + \epsilon_1 \end{cases}$   
Best lift  $\leftarrow$

# Oblique shocks + expansion waves [10.1]

flat plate at angle  $\alpha$



Total force/unit depth

$$R' = (p_3 - p_2)c$$

$$\Rightarrow L' = (p_3 - p_2)c \cos \alpha$$

$$L = (p_L - p_u)c \cos \alpha$$

$$D' = (p_3 - p_2)c \sin \alpha \neq 0$$

$$D = (p_L - p_u)c \sin \alpha$$

*shock wave drag  $D' \neq 0$*

$p_2$  and  $p_3$  can be found

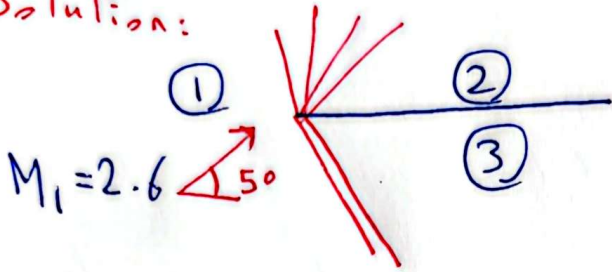
$$\Rightarrow \frac{L}{D} = \frac{C_L}{C_D} = \frac{\cos \alpha}{\sin \alpha}$$

$$E_x.$$

$M_1 = 2.6 \Delta \alpha$

Find  $c_L, c_d$  for  $\alpha = 5^\circ$   
 $15^\circ$   
 $30^\circ$

Solution:



1-2 : E.W

turn angle:  $\alpha = 5^\circ$

$$M_1 = 2.6, \alpha = 5^\circ \xrightarrow[\text{Calc}]{\text{online}} \nu_1 = 41.41^\circ$$

$$v_2 = v_1 + \alpha = 41.41 + 5 = 46.41 \xrightarrow[\text{Calc}]{\text{online}} \begin{cases} M_2 = 2.83 \\ \frac{P_2 t_2}{P_1} = 28.43 \end{cases}$$

$$\frac{P_2}{P_1} = \frac{P_2}{P_{t_2}} \times \frac{P_{t_2}}{P_{t_1}} \times \frac{P_{t_1}}{P_1} = \frac{1}{28.43} \times 1 \times 19.95 = 0.7022 < 1$$

$$1-3: 0.5 \quad M_1 = 2.6 \quad \delta = \alpha = 5^\circ \longrightarrow \begin{matrix} 0.5 \\ \text{calc} \end{matrix} \left\{ \begin{array}{l} \theta = 26.5^\circ \\ \frac{P_3}{P_1} = 1.403 \end{array} \right.$$

$$C_\theta = \frac{2}{\gamma M_1^2} \left( \frac{P_3}{P_1} - \frac{P_2}{P_1} \right) \cos \alpha = 0.148$$

$$C_d = \quad \quad \quad \sin \alpha = 0.0129$$

other angles of attack are similar

Example:  $M_1 = 2.6$  evaluate  $C_\ell$ ,  $C_d$  for  $5^\circ$ ,  $15^\circ$ ,  $30^\circ$

for  $M_1 = 2.6$ ,  $\delta = 5^\circ$  from  $\delta$ - $\theta$ - $M$  diagram  $\theta = 26.5^\circ$

$$M_{n_1} = M_1 \sin \theta = 2.6 \times \sin 26.5^\circ =$$

from normal shock tables: for  $M_{n_1} = 1.16$   $\frac{p_3}{p_1} = 1.403 > 1$

from isentropic tables:  $M_1 = 2.6$   $\frac{p_{t1}}{p_1} = 19.95$

from expansion tables: for  $M_1 = 2.6$   $\nu_1 = 41.41^\circ$

$$\nu_2 = \nu_1 + \theta = 41.41 + 5 = 46.41^\circ \Rightarrow M_2 = 2.83 \quad (M_2 > M_1)$$

from isentropic tables: for  $M_2 = 2.83$   $\frac{p_{t2}}{p_2} = 28.43$

expansion fan  
( $p_{t1} = p_{t2}$  isentropic)  $\frac{p_2}{p_1} = \frac{p_2}{p_{t2}} \frac{p_{t2}}{p_{t1}} \frac{p_{t1}}{p_1} = \frac{1}{28.43} \cdot 1 \cdot (19.95) = 0.7022$

$$q_\infty = \frac{1}{2} \gamma p_\infty M_\infty^2 \quad C_l = \frac{2}{\gamma M_1^2} \left( \frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \cos \alpha = 0.148$$

$$C_d = \frac{2}{\gamma M_1^2} \left( \frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \sin \alpha = C_l \tan \alpha = 0.0129$$

$$\alpha = 5^\circ \rightarrow C_\ell = 0.148 \quad C_d = 0.0129$$

$$\alpha = 15^\circ \rightarrow C_\ell = 0.452 \quad C_d = 0.121$$

$$\alpha = 30^\circ \rightarrow C_\ell = 1.19 \quad C_d = 0.687$$